

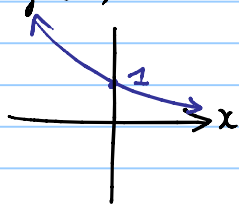
Unit 1.7 Exponential Functions (pp. 45-53)

Note Title

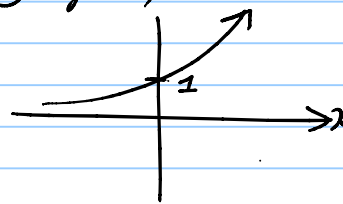
11/2/2020

Exponential function: $y = a^x$ with $a > 0, a \neq 1$.

① $y = a^x, 0 < a < 1$



② $y = a^x, a > 1$

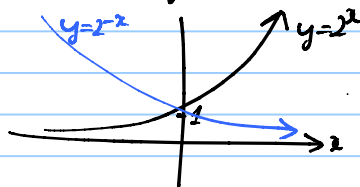


When $a = 1$, $y = a^x = y = 1$.

Examples.

1) See Fig 3 on p. 47.

2) $y = 2^x$ and $y = (\frac{1}{2})^x = 2^{-x}$



The number e (Euler's number) - p. 50.

The natural exponential function: $y = e^x$.

Examples:

1) Sketch

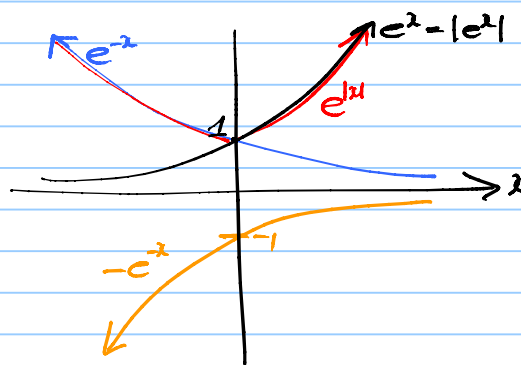
$y = e^x$

$y = e^{-x}$

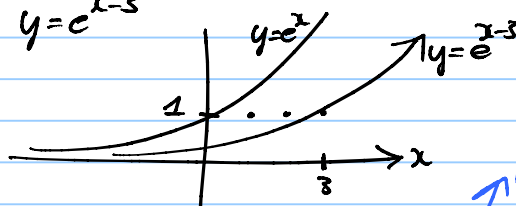
$y = -e^{-x}$

$y = e^{1/x}$

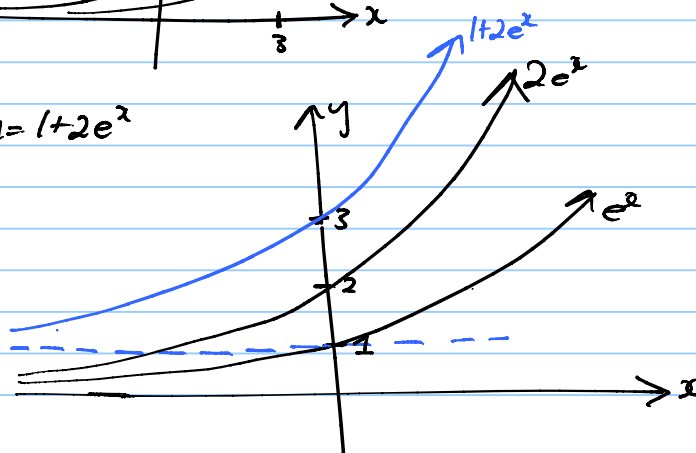
$y = |e^x|$



② $y = e^{x-3}$



③ $y = 1 + 2e^x$



Laws of Exponents (p. 47).

Examples:

$$1) \frac{3^x (3^x)^6}{3^{4x}} = \frac{3^x \cdot 3^{6x}}{3^{4x}} = 3^{x+6x-4x} = 3^x$$

$$2) \frac{e^3 \sqrt{e^4}}{e^8} = \frac{e^3 \times e^{4/2}}{e^8} = \frac{e^3 \cdot e^2}{e^8} = e^{3+2-8} = e^{-3} = \frac{1}{e^3}$$

3) Find the domain of $g(t) = \sin(e^t)$

Suppose $f(t) = e^t$ and $h(t) = \sin(t)$

$$D_f = \mathbb{R}, \quad R_f = \mathbb{R}$$

$$D_h = \mathbb{R}$$

$$\Rightarrow D_g = \mathbb{R}$$

(4) Find domain of $f(x) = \frac{1}{1+e^x}$

$$1+e^x \neq 0 \Rightarrow e^x \neq -1, \text{ but } e^x + 1 > 0 \forall x \in \mathbb{R} \\ \Rightarrow D_f = \mathbb{R}.$$